The Pais–Uhlenbeck Metric as a Minimum-Distortion Principle, and the Representation Problem for the Fourth-Order Field

GPT-5.6.sol* Asger Alstrup Palm†

Open-repository preprint, July 2026

Abstract

For the unequal-frequency PT-symmetric Pais–Uhlenbeck oscillator, the admissible complex-symplectic diagonalizers form a coset $S_+\, SO(2,\mathbb{C})^2$ of the unique Hermitian positive-definite diagonalizer S_+. We prove that S_+ minimizes the affine-invariant metric distortion: for every admissible S, $\|\log(S^\dagger S)\|_F^2 \geq 4r^2$ with $r = \log\frac{\omega_1+\omega_2}{\omega_1-\omega_2}$, with equality exactly on the unitary stabilizer coset, on which the induced Hilbert-space metric is constantly the Bender–Mannheim $\eta = e^{-Q}$. The proof is elementary and exact: at worst-case stabilizer alignment the log-eigenvalue pair of $S^\dagger S$ is $\{\operatorname{arccosh}(\cosh r \cosh b) + a,\ |\operatorname{arccosh}(\cosh r \cosh b) - a|\}$ in the Gram rapidities (a, b) of the stabilizer factor. The general result behind it is an orthogonal Cartan-norm projection principle, proved in if-and-only-if form, whose principal case is the $\mathrm{Sp}(2n,\mathbb{C})$ inter-mode theorem: relative to any symplectic mode splitting, minimality along the mode-preserving coset holds exactly for inter-mode diagonalizer directions. The Pais–Uhlenbeck oscillator is the fully integrable $n = 2$ model: the physical stabilizer, itself not θ-stable, has canonical compatible hull the mode-preserving $SL(2,\mathbb{C})^2$ (totally geodesic $\mathbb{H}^3 \times \mathbb{H}^3$; generically, with the alignment stratification made explicit), the Bender–Mannheim direction is inter-mode, and the closed form above is the resulting mixed hyperbolic/flat Pythagorean law. We then quantize the free fourth-order field modewise and compute the Bender–Mannheim ground state exactly: it is a normalizable real Gaussian for every mode — indeed exactly the Dyson pullback $\rho^{-1}\phi_0$ of the normal-form vacuum, so the modewise Dyson maps are *pointed* η-unitaries and the physical PT completion is globally unitarily equivalent to the positive normal-form Fock completion. If instead the two Gaussian vacua are compared through the identity embedding of the auxiliary canonical variables and the standard involution, their identity-embedding Gaussian fidelity tends to the universal constant $\sqrt{3}/2$ at large momentum, their auxiliary standard-Fock occupation to $1/3$ per mode pair, and the two auxiliary standard-CCR product states are disjoint in every spatial dimension. The obstruction is thus not physical Hilbert-space inequivalence but the incompatibility between the identity embedding of canonical variables and the Dyson-transported physical involution. All identities are machine-verified.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the correspond-

*OpenAI model and principal programme author.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. asger@area9.dk.

ing human contact. The project is an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

1 Introduction

This paper continues the symplectic analysis of the PT-symmetric Pais–Uhlenbeck (PU) oscillator begun in [1], whose results and conventions we use freely: $V = (x, y, p, q)^\top$, $[V_a, V_b] = iJ_{ab}$, $H_{\mathrm{PT}} = \frac{1}{2}V^\top G V$ with the PU matrix G, positive normal form G_0, rapidity

$$r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}, \qquad \omega_1 > \omega_2 > 0, \tag{1}$$

and, after the canonical normalization $d_x d_y = \gamma\omega_1\omega_2$, the unique Hermitian positive-definite diagonalizer

$$S_+ = \cosh \tfrac{r}{2}\, I + \sinh \tfrac{r}{2}\, B \in \mathrm{Sp}(4, \mathbb{C}), \qquad S_+^\top G' S_+ = G_0', \qquad S_+^\top J S_+ = J, \tag{2}$$

with B a fixed Hermitian involution. The full solution family of the diagonalization problem is the coset $S_+\,\mathrm{Stab}(J, G_0')$ with $\mathrm{Stab}(J, G_0') \cong SO(2, \mathbb{C}) \times SO(2, \mathbb{C})$, and the metaplectic logarithm of S_+ is the Bender–Mannheim generator $Q = \alpha pq + \beta xy$ [3, 4, 1].

Two questions are addressed here.

First, uniqueness of the positive diagonalizer does not by itself distinguish S_+ variationally: one may ask whether S_+ *minimizes* a natural functional over all admissible diagonalizers. We prove that it minimizes the affine-invariant distortion $\mathcal{F}(S) = d(I, S^\dagger S)^2 = \|\log(S^\dagger S)\|_F^2$ (Theorem 2.5), so that the Bender–Mannheim metric is selected by a minimum-distortion principle: it is the least-distorting complex canonical transformation converting the PU Hamiltonian to its positive normal form. The results of Section 2 form a hierarchy:

normal-coset projection principle (Thm. 2.9) $\Longrightarrow$ $\mathrm{Sp}(2n, \mathbb{C})$ inter-mode theorem (Thm. 2.14) $\Longrightarrow$ exact $\mathbb{H}^2\times$

The first is classical symmetric-space geometry, stated here in if-and-only-if form; the second is the principal general result, making the selection principle a theorem for arbitrary quadratic PT mode systems; the third is what only the integrable $n = 2$ case supplies — the exact minimizing value, obtained by the elementary two-invariant derivation that opens the section. The physical stabilizer itself fails θ-stability — which is why the classical theory does not apply to it directly — but its canonical compatible hull is exactly the mode-preserving $SL(2, \mathbb{C})^2$ (Proposition 2.10, stratified over the alignment locus).

Second, for the free fourth-order field $(\Box + m_1^2)(\Box + m_2^2)\phi = 0$ each spatial momentum k contributes a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$, and the modewise construction applies with rapidity

$$r(k) = \log \frac{\big(\omega_1(k) + \omega_2(k)\big)^2}{m_1^2 - m_2^2} = 2\log|k| + O(1) \qquad (m_1 > m_2). \tag{3}$$

One might expect from (3) that the modewise transformation carries unboundedly growing squeezing and fails Fock implementability "dramatically". The exact computation shows something sharper and more structured (Theorems 3.2–3.5 and Proposition 3.4): the PT ground state exists as a normalizable Gaussian for *every* mode, and it is exactly the Dyson pull-back $\psi_0 = \rho^{-1}\phi_0$ of the normal-form vacuum. Two comparisons must then be sharply distinguished. *Physical completion*: the modewise Dyson maps are pointed η-unitaries, their pointed infinite tensor product exists, and the physical PT completion is globally unitarily equivalent to the positive normal-form Fock completion (Proposition 3.4). *Auxiliary identity embedding*: if instead both Gaussian vacua are regarded as vectors of one standard L^2 space, with the auxiliary canonical variables identified by the identity map and the standard involution retained, their mismatch saturates at *universal constants* — identity-embedding fidelity $\sqrt{3}/2$ and auxiliary standard-Fock occupation $1/3$ per mode pair at large k — and the two auxiliary standard-CCR product states are disjoint in every spatial dimension $d \geq 1$, with overlap $|\langle\Omega, \Psi\rangle|^2 = \exp[-(\log \frac{2}{\sqrt{3}}) \frac{\mathrm{vol}(B_d)}{(2\pi)^d} V\Lambda^d (1 + o(1))]$ under a cutoff. The obstruction is the incompatibility of the identity embedding with the Dyson-transported involution, not physical Hilbert-space inequivalence. The divergence of $r(k)$ itself is a statement about the observable-space (phase-space) metric, which we separate cleanly (Corollary 3.1): in the canonically normalized mode bundle its eigenvalues $e^{\pm r(k)} \asymp (1 + k^2)^{\pm 1}$ give the two spectral directions relative weights of orders $+2$ and -2 — a Sobolev-pair structure relative to that trivialization — while the state-level mismatch remains $O(1)$ per mode.

Verification. As in [1], every finite-dimensional identity in this paper — the invariant formulas, the closed-form log-eigenvalues (including the absolute-value form of (8)), the PT ground-state Gaussian and its Dyson-pullback identity (23), the eigenvalue equation, the fidelity and occupation formulas, and the limits $\sqrt{3}/2$ and $1/3$ — is machine-verified symbolically, with independent numerical checks (global optimization for Theorem 2.5; high-precision quadrature for Theorem 3.2). The artifact is the root of the `area9innovation/weyl-gravity` repository of [1] (scripts `symbolic/verify_variational_fock.py`, `symbolic/verify_paper2_referee.py`, `numeric/distortion_scan.py`, `numeric/cartan_checks.py`; SymPy 1.14.0, Python 3.14; the release tag `paper2-v1.3` fixes the commit).

2 The minimum-distortion theorem

Work in the canonically normalized coordinates of [1]; all adjoints refer to them. The stabilizer is parametrized by two complex angles,

$$C(\theta_1, \theta_2) = \sum_{j=1,2} \big(\cos\theta_j\, P_j + \sin\theta_j\, X_j \big), \qquad X_j^2 = -P_j, \tag{4}$$

with P_j the projections onto the modes of G_0' and X_j the per-mode complex structures; every admissible diagonalizer is $S = S_+ C$.

Definition 2.1. For an admissible S let $\mathcal{F}(S) = \|\log(S^\dagger S)\|_F^2$, the squared affine-invariant distance of the observable-space metric $S^\dagger S$ from the identity [5]. For a stabilizer element $C = C_1 \oplus C_2$ (blocks on the two modes) define its *Gram rapidities* $\tau_j \geq 0$ by

$$W_j = C_j C_j^\dagger, \qquad \det W_j = 1, \qquad \cosh\tau_j = \tfrac{1}{2}\operatorname{tr} W_j, \tag{5}$$

and set $a = \frac{1}{2}(\tau_1 + \tau_2)$, $b = \frac{1}{2}(\tau_1 - \tau_2)$.

Note $\tau_1 = \tau_2 = 0$ iff $W_1 = W_2 = I$ iff C is unitary. Since $P = S^\dagger S = C^\dagger S_+^2 C$ is Hermitian positive *and* symplectic, its eigenvalues form reciprocal pairs $e^{\pm x_1}, e^{\pm x_2}$ with $x_i \geq 0$, and $\mathcal{F}(S) = 2(x_1^2 + x_2^2)$.

Lemma 2.2 (Invariants)**.** *Write $W = CC^\dagger = W_1 \oplus W_2$, $c = \cosh r$, $s = \sinh r$. Decomposing $W_j = \cosh\tau_j\, I + \sinh\tau_j\,(n_j \cdot \sigma)$ with real unit vectors n_j, and letting $\chi \in [-1, 1]$ denote the alignment $n_1 \cdot \tilde{n}_2$ (where $\tilde{n}_2$ is n_2 with its σ_2 component reflected),*

$$\cosh x_1 + \cosh x_2 = c\,(\cosh\tau_1 + \cosh\tau_2), \tag{6}$$

$$\cosh x_1 \cosh x_2 = \tfrac{1}{2}\Big[(2c^2 - s^2)\cosh\tau_1\cosh\tau_2 + s^2\big(1 + \sinh\tau_1\sinh\tau_2\,\chi\big)\Big]. \tag{7}$$

Proof. $\operatorname{tr} P = \operatorname{tr}(e^{rB}W) = c\operatorname{tr} W + s\operatorname{tr}(BW)$, and $\operatorname{tr}(BW) = 0$ because B is off-diagonal and W diagonal with respect to the mode splitting; this gives (6). For (7), expand $\operatorname{tr} P^2 = c^2\operatorname{tr} W^2 + s^2\operatorname{tr}\big((BW)^2\big)$ (the cross terms again vanish blockwise); the off-diagonal blocks of B equal $-\sigma_2$, and the 2×2 identity $\sigma_2 M \sigma_2 = (\det M)\,M^{-\top}$ reduces $\operatorname{tr}\big((BW)^2\big) = 2\operatorname{tr}(\overline{W_2}^{-1}W_1) = 4\big[\cosh\tau_1\cosh\tau_2 - \sinh\tau_1\sinh\tau_2\,\chi\big]$. Converting $(\operatorname{tr} P, \operatorname{tr} P^2)$ to the symmetric functions of $\cosh x_i$ yields (7). □

(As verification evidence, both matrix identities are also checked against random stabilizer elements in the repository.)

Lemma 2.3 (Worst alignment; closed form)**.** *At $\chi = -1$ the system (6)–(7) is solved exactly by the unordered nonnegative pair*

$$\{x_1, x_2\} = \big\{\varphi(b) + a,\ \big|\varphi(b) - a\big|\big\}, \qquad \varphi(b) = \operatorname{arccosh}\big(\cosh r\,\cosh b\big)\ \geq\ r \tag{8}$$

(the absolute value is required: for $a > \varphi(b)$ the signed expression $\varphi - a$ is negative, while the x_i are defined as nonnegative log-eigenvalue exponents; equivalently $\cosh x_{1,2} = \cosh(\varphi \pm a)$ with the principal nonnegative arccosh*).*

Proof. With $\alpha = \cosh a$, $\beta = \cosh b$ one finds from (6)–(7) at $\chi = -1$, using $\cosh\tau_1\cosh\tau_2 \pm \sinh\tau_1\sinh\tau_2 = \cosh(\tau_1 \pm \tau_2)$,

$$\cosh x_1 + \cosh x_2 = 2c\,\alpha\beta, \qquad \cosh x_1\cosh x_2 = \alpha^2 + c^2\beta^2 - 1,$$

whence $\cosh x_{1,2} = c\alpha\beta \pm \sqrt{(\alpha^2 - 1)(c^2\beta^2 - 1)} = \cosh(\varphi \pm a)$ with $\cosh\varphi = c\beta$, by the hyperbolic addition formulas (elementary identities; also machine-checked in the repository as verification evidence); since cosh is even, the nonnegative exponents are $\varphi + a$ and $|\varphi - a|$. Finally $\cosh\varphi(b) - \cosh r = \cosh r(\cosh b - 1) \geq 0$ gives $\varphi(b) \geq r$ with equality iff $b = 0$. □

Lemma 2.4 (Alignment monotonicity)**.** *At fixed $\cosh x_1 + \cosh x_2$, the quantity $x_1^2 + x_2^2$ is nondecreasing in $\cosh x_1 \cosh x_2$; and (7) is nondecreasing in χ.*

Proof. Parametrize $\cosh x_{1,2} = T/2 \pm w$, $w \geq 0$; the product decreases in w, and $\frac{d}{dw}(x_1^2 + x_2^2) = 2\big(\frac{x_1}{\sinh x_1} - \frac{x_2}{\sinh x_2}\big) \leq 0$ because $x/\sinh x$ is strictly decreasing and $x_1 \geq x_2$. The second statement is the sign of the coefficient $s^2\sinh\tau_1\sinh\tau_2/2 \geq 0$ in (7). □

Theorem 2.5 (Minimum distortion). *For every admissible diagonalizer $S = S_+C$,*

$$\mathcal{F}(S) \;\geq\; 4\Big[a^2 + \operatorname{arccosh}^2\big(\cosh r \cosh b\big)\Big] \;\geq\; 4r^2 = \mathcal{F}(S_+), \tag{9}$$

with overall equality if and only if C is unitary. All minimizers share the polar factor S_+, hence the same Hilbert-space metric $\eta = e^{-Q}$: the Bender–Mannheim metric is the unique metric of any minimum-distortion diagonalizer,

$$S_+ \in \operatorname*{arg\,min}_{S^\top JS=J,\ S^\top G'S=G'_0} d\big(I, S^\dagger S\big), \qquad Q = -2\log\big(\textit{polar factor of any minimizer}\big)\big|_{\textit{metaplectic}}. \tag{10}$$

Proof. By Lemma 2.4, lowering χ to -1 at fixed Gram rapidities does not increase $x_1^2 + x_2^2$; by Lemma 2.3 the value there is $\mathcal{F} = 2\big[(\varphi+a)^2 + (\varphi-a)^2\big] = 4(\varphi^2 + a^2)$; and $\varphi(b) \geq r$. Equality in the chain forces $a = 0$ and $b = 0$, i.e. $\tau_1 = \tau_2 = 0$, i.e. C unitary; the converse is immediate since then $S^\dagger S$ is unitarily conjugate to S_+^2. If C is unitary, $S_+C = S_+ \cdot C$ is already a polar decomposition, so the polar factor is S_+ and the metaplectic implementer factorizes as $\rho' = U_C\, e^{-Q/2}$ with U_C unitary, giving $\eta' = \rho'^\dagger\rho' = e^{-Q}$ for every minimizer. □

Remark 2.6. (i) The individual log-eigenvalues do *not* satisfy $x_i \geq r$: numerical scans show $\min_i x_i < r$ on open regions of the stabilizer; only the joint quantity is bounded below, which is why the two-invariant argument is needed. (ii) The equality set is the unitary part of the stabilizer, which in the canonical coordinates is generically the finite group $\mathbb{Z}_2 \times \mathbb{Z}_2$ [1]; abstractly it is contained in the maximal compact $U(1)^2$ of $SO(2,\mathbb{C})^2$. (iii) The theorem is an instance of an orthogonal Cartan-norm projection principle, proved in Section 2.1 together with its converse; the elementary proof above remains valuable because it yields the exact value (8), which the geometric argument does not.

2.1 Geometric recognition: an orthogonal Cartan-norm projection principle

Normalization dictionary. Define the Cartan projection $\mu(S)$ of $S \in \mathrm{Sp}(4,\mathbb{C})$ as the chamber-ordered log-spectrum of $S^\dagger S$ (twice the usual log-singular-value vector), equip the Hermitian directions $\mathfrak{p}$ with the trace form $\langle \xi, \eta\rangle = \operatorname{tr}(\xi\eta)$, and normalize the symmetric-space distance by $d_X(o, e^\xi \cdot o) = \|\xi\|_F$. If $\operatorname{spec}(\xi) = \{\pm a_1, \pm a_2\}$ then $\mu(e^\xi) = (2a_1, 2a_2)$ (chamber-ordered), $\|\xi\|_F^2 = 2\sum_i a_i^2$, and the three minimands used below are related by

$$\boxed{\mathcal{F}(g) = 2\|\mu(g)\|^2, \qquad \mathcal{F}(e^\xi) = 4\, d_X(o, e^\xi \cdot o)^2 = \|2\xi\|_F^2.} \tag{11}$$

In particular $\min\sqrt{\mathcal{F}} = \|2\xi\|_F$, $\min\|\mu\| = \|2\xi\|_F/\sqrt{2}$, and the first-variation coefficient 8 in (13) below is consistent with $\mathcal{F}(e^\xi) = 4\|\xi\|_F^2$. Throughout, K denotes the maximal compact subgroup determined by the physical inner product: $K = \mathrm{Sp}(4,\mathbb{C}) \cap U(4) \cong \mathrm{Sp}(2)$ (the compact symplectic group, written $USp(4)$ in part of the literature), and in the n-mode setting $K = \mathrm{Sp}(2n,\mathbb{C}) \cap U(2n) \cong \mathrm{Sp}(n)$. With these conventions $\mu(S_+) = (r,r)$, which lies *on the chamber wall* $x_1 = x_2$ of type C_2.

Theorem 2.5 is thus a minimization of the Cartan-projection *norm* along a stabilizer coset. We emphasize at the outset what is classical and what is not. The scalar projection inequality below is standard symmetric-space geometry in the tradition of Mostow's self-adjoint groups [6, 8]: the compatible-subgroup orbit is totally geodesic and convex, and normal geodesics realize nearest-point projection. (Kempf–Ness theory [7] offers a related variational perspective on minimal-norm points of complexified orbits, but it is not the direct source of this coset inequality.) The inequality controls $\|\mu\|$, not the vector-valued Cartan projection in Weyl-chamber order, and we do not claim it as new. The structure specific to this problem is: (i) the physically given stabilizer $H = \mathrm{Stab}(J, G'_0)$ is *not* θ-stable, so the classical theory does not see it directly; (ii) its canonical θ-stable hull is exactly the mode-preserving $SL(2,\mathbb{C}) \times SL(2,\mathbb{C})$ (Proposition 2.10), through which minimality is inherited by restriction; and (iii) the exact $\mathbb{H}^2 \times \mathbb{R}$ value formula (8).

Let $\mathbb{P}$ denote the cone of positive-definite Hermitian matrices with the affine-invariant metric, a nonpositively curved (NPC) space in which $t \mapsto d(\gamma(t), x)^2$ is strictly convex along nonconstant geodesics, with $\frac{d}{dt} d(e^{t\zeta}, P)^2\big|_{t=0} = -2\langle \zeta, \log P\rangle$ [5, 8].

Definition 2.7 (Compatible subgroup)**.** A closed subgroup $\mathcal{C} \subset GL(n,\mathbb{C})$ is *compatible* if it is closed under conjugate transpose and admits the global Cartan decomposition

$$\mathcal{C} = K_{\mathcal{C}} \exp \mathfrak{c}_{\mathfrak{p}}, \qquad K_{\mathcal{C}} = \mathcal{C} \cap U(n), \qquad \mathfrak{c}_{\mathfrak{p}} = \{\eta = \eta^\dagger : e^{t\eta} \in \mathcal{C}\ \forall t \in \mathbb{R}\}. \tag{12}$$

Every connected closed $\dagger$-stable reductive matrix group is compatible; mere closedness under $\dagger$ is *not* sufficient: the discrete group $\mathcal{C} = \{A^n : n \in \mathbb{Z}\}$, $A = \mathrm{diag}(2, \frac{1}{2}) \in \mathrm{Sp}(2,\mathbb{R})$, is closed and self-adjoint with all elements positive, yet $A^{1/2} \notin \mathcal{C}$ and $\mathfrak{c}_{\mathfrak{p}} = 0$, and it violates the conclusions of Lemma 2.8 and Theorem 2.9 below (take $\xi = -\frac{3}{4}\log A$: then $c = A$ beats $c = e$). Compatibility excludes such counterexamples.

Lemma 2.8. *Let $\mathcal{C} \subset GL(n,\mathbb{C})$ be a compatible subgroup and let $\mathbb{P}_{\mathcal{C}} = \mathcal{C} \cap \mathbb{P}$ be its set of positive elements. Then $\mathbb{P}_{\mathcal{C}} = \exp \mathfrak{c}_{\mathfrak{p}}$ is a topologically closed, geodesically closed (hence convex) subset of $\mathbb{P}$ — so it admits unique metric projection from any point of $\mathbb{P}$ — and $\mathbb{P}_{\mathcal{C}} = \{cc^\dagger : c \in \mathcal{C}\}$.*

Proof. A positive element of $\mathcal{C}$ has, by the global decomposition and uniqueness of the polar decomposition, trivial unitary part, so $\mathbb{P}_{\mathcal{C}} = \exp \mathfrak{c}_{\mathfrak{p}}$; in particular $A \in \mathbb{P}_{\mathcal{C}}$ has $\log A \in \mathfrak{c}_{\mathfrak{p}}$ and $A^t = e^{t \log A} \in \mathcal{C}$ for all real t. For $A, B \in \mathbb{P}_{\mathcal{C}}$ the geodesic is $t \mapsto A^{1/2}\big(A^{-1/2} B A^{-1/2}\big)^t A^{1/2}$ [5]; it lies in $\mathcal{C}$ because $A^{\pm 1/2}$, the positive matrix $M = A^{-1/2} B A^{-1/2} \in \mathbb{P}_{\mathcal{C}}$, and its real powers M^t all do, and it consists of positive elements, hence stays in $\mathbb{P}_{\mathcal{C}}$. For the Gram-set claim: $cc^\dagger = k\, e^{2\eta} k^{-1} = e^{2\,\mathrm{Ad}_k \eta} \in \exp \mathfrak{c}_{\mathfrak{p}}$ for $c = ke^{\eta}$ ($\mathrm{Ad}_{K_{\mathcal{C}}}$ preserves $\mathfrak{c}_{\mathfrak{p}}$), and conversely $A = (A^{1/2})(A^{1/2})^\dagger$ with $A^{1/2} \in \mathcal{C}$. Topological closedness of $\mathbb{P}_{\mathcal{C}} = \mathcal{C} \cap \mathbb{P}$ is immediate from closedness of $\mathcal{C}$; unique metric projection onto closed convex subsets is standard in NPC spaces [8]. ☐

Theorem 2.9 (Orthogonal Cartan-norm projection principle)**.** *Let $\mathcal{C} \subset GL(n,\mathbb{C})$ be a compatible subgroup (Definition 2.7) and let $\xi = \xi^\dagger$. Then:*

(a) For every $\eta \in \mathfrak{c}_{\mathfrak{p}}$,

$$\frac{d}{dt}\Big|_{t=0} \big\| \log\big((e^{\xi} e^{t\eta})^\dagger e^{\xi} e^{t\eta}\big) \big\|_F^2 = 8\, \langle \eta, \xi \rangle. \tag{13}$$

(b) If $\xi \perp \mathfrak{c}_{\mathfrak{p}}$, then for every $c \in \mathcal{C}$,

$$\big\| \log\big((e^{\xi} c)^\dagger e^{\xi} c\big) \big\|_F \ \geq\ \|2\xi\|_F, \qquad \operatorname*{arg\,min}_{c \in \mathcal{C}} \big\| \log\big((e^{\xi} c)^\dagger e^{\xi} c\big) \big\|_F = \mathcal{C} \cap U(n). \tag{14}$$

(c) *Conversely, if the tangential part* $\xi_T = \operatorname{proj}_{\mathfrak{c}_\mathfrak{p}} \xi$ *is nonzero, then* $c = e$ *is not a minimizer: by* (13) *with* $\eta = -\xi_T$ *the functional decreases at rate* $8\|\xi_T\|^2$. *Hence*

$$e \text{ minimizes } c \mapsto \left\| \log\left((e^{\xi}c)^\dagger e^{\xi}c\right)\right\|_F \text{ on } \mathcal{C} \quad \Longleftrightarrow \quad \xi \perp \mathfrak{c}_\mathfrak{p}. \tag{15}$$

Proof. By congruence invariance of the affine metric, $d\big(I, (e^{\xi}c)^\dagger e^{\xi}c\big) = d\big((cc^\dagger)^{-1}, e^{2\xi}\big)$, and $(cc^\dagger)^{-1} = dd^\dagger$ with $d = c^{-\dagger} \in \mathcal{C}$, so it ranges over $\mathbb{P}_\mathcal{C}$ as c ranges over $\mathcal{C}$ (Lemma 2.8). For (a), take $c = e^{t\eta}$: then $(cc^\dagger)^{-1} = e^{-2t\eta}$ is a geodesic through I with velocity -2η, and the NPC first-variation formula gives $\frac{d}{dt} d(e^{-2t\eta}, e^{2\xi})^2\big|_0 = -2\langle -2\eta, 2\xi\rangle = 8\langle \eta, \xi\rangle$. For (b), every point of $\mathbb{P}_\mathcal{C}$ is joined to I by a geodesic $t \mapsto e^{2t\eta}$, $\eta \in \mathfrak{c}_\mathfrak{p}$, inside $\mathbb{P}_\mathcal{C}$; along it $h(M) = d(M, e^{2\xi})^2$ is strictly convex unless constant, and its first variation at I vanishes by orthogonality. Hence I is the unique minimizer of h on $\mathbb{P}_\mathcal{C}$, with value $\|2\xi\|_F^2$; the minimizing c are exactly those with $cc^\dagger = I$. Part (c) is immediate from (a). $\square$

The numerical experiments that preceded this proof (random tangential and normal directions ξ) are retained in the repository as regression checks of (13) and of both directions of (c).

Proposition 2.10 (Canonical compatible hull, stratified)**.** *Let* $\theta(g) = (g^\dagger)^{-1}$ *and let* $C_\theta(H)$ *be the smallest closed subgroup containing* $H = \operatorname{Stab}(J, G_0')$ *and* $\theta(H)$. *In the canonical coordinates,*

$$C_\theta(H) = C_1 \times C_2, \qquad C_j = \begin{cases} SL(2,\mathbb{C}), & \lambda_j \neq 1, \\ SO(2,\mathbb{C}), & \lambda_j = 1, \end{cases} \tag{16}$$

acting on the j*-th mode plane; here* $\lambda_j > 0$ *is the dimensionless mode-aspect parameter of the canonical coordinates ([1];* $\lambda_1 = \omega_2$, $\lambda_2 = \omega_1$ *in normalized units for the symmetric split, with the split-invariant* $\lambda_1/\lambda_2 = \omega_2/\omega_1 < 1$*). On the generic stratum the compatible hull is the full mode-preserving symplectic subgroup: not merely a convenient supergroup but the minimal* θ*-stable subgroup forced by the physical stabilizer. At an alignment value* $\lambda_j = 1$ *the* j*-th factor of* H *is already* θ*-stable and the hull degenerates to* H_j *itself.*

Proof. H and $\theta(H)$ are contained in the block-diagonal subgroup, so $C_\theta(H) \subseteq SL(2,\mathbb{C})^2$. For the converse it suffices to show the real Lie algebra generated by $\mathfrak{h}_j = \mathbb{C}X_j$ and $\theta(\mathfrak{h}_j) = \mathbb{C}X_j^\dagger$ is all of $\mathfrak{sl}(2,\mathbb{C})$ on each mode. On mode 1 (block coordinates), $X_1 = \left(\begin{smallmatrix} 0 & \lambda_1 \\ -\lambda_1^{-1} & 0\end{smallmatrix}\right)$ (with $\lambda_1 = \omega_2$ in normalized units for the symmetric split) and

$$[X_1, X_1^\dagger] = (\lambda_1^2 - \lambda_1^{-2}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0 \qquad (\lambda_1 \neq 1), \tag{17}$$

and the real span of $X_1, X_1^\dagger, [X_1, X_1^\dagger]$ and their i-multiples is six-dimensional, hence equals $\mathfrak{sl}(2,\mathbb{C})$; likewise on mode 2 with λ_2. The generated group is then the connected $SL(2,\mathbb{C})$ on each generic mode. At $\lambda_j = 1$ one has $X_j^\dagger = -X_j$, so $\theta(\mathfrak{h}_j) = \mathfrak{h}_j$ and the closure is $\mathfrak{so}(2,\mathbb{C})$ itself. $\square$

(As regression evidence, not as part of the proof: the Lie-closure dimensions 6 and 2 are also confirmed computationally in the repository.)

Remark 2.11 (Alignment: hull contraction versus physical enhancement). The stratification of Proposition 2.10 is mirrored by two distinct discontinuities at the alignment locus $\lambda_j = 1$, which must not be conflated. For the *hull*, the compact locus contracts, $C_j \cap K: \ SU(2) \to U(1)$, because the hull itself shrinks from $SL(2,\mathbb{C})$ to $SO(2,\mathbb{C})$; accordingly, for minimization over the hull the equality locus *changes* from $SU(2)$-type to $U(1)$-type. For the *physical stabilizer* the unitary locus is enhanced, $H_j \cap K: \ \mathbb{Z}_2 \to U(1)$, and it is only for the variational problem restricted to H that the equality locus of Theorem 2.5 *enlarges* at alignment. The enhancement is a property of the physical embedding (the twist), not of the abstract groups, and marks the parameter values at which the twisted and compatible pictures coincide.

Corollary 2.12 (Second proof of Theorem 2.5)**.** *Apply Theorem 2.9(b) with $\mathcal{C}$ the stratified compatible hull $C_\theta(H)$ of Proposition 2.10: on the generic stratum $\mathcal{C} = SL(2,\mathbb{C})^2$ — connected, †-stable and reductive, hence compatible — and at alignment one or both factors reduce to $SO(2,\mathbb{C})$; orthogonality of ξ holds on every stratum because ξ is block-off-diagonal. On the generic stratum the positive part $\mathbb{P}_{\mathcal{C}} \cong \mathbb{H}^3 \times \mathbb{H}^3$ is totally geodesic in $\mathbb{P}$, the Gram data (τ_j, n_j) of Lemma 2.2 being polar coordinates on the two hyperbolic factors; take $\xi = \frac{r}{2}B$, which is block-off-diagonal and hence orthogonal to $\mathfrak{c}_\mathfrak{p}$ (the per-mode traceless Hermitian blocks). This yields $\mathcal{F}(S_+c) \geq \mathcal{F}(S_+)$ for every c in the hull, with the two equality records kept separate:*

$$\arg\min_{c\in\mathcal{C}} \mathcal{F}(S_+c) = \mathcal{C} \cap K, \qquad \arg\min_{h\in H} \mathcal{F}(S_+h) = H \cap K, \tag{18}$$

the second following from the first because $e \in H \subset \mathcal{C}$ realizes the unrestricted minimum.

Remark 2.13. (i) The closed form (8) is the Pythagorean law of the ambient symmetric space: the worst-alignment configurations sweep a totally geodesic $\mathbb{H}^2 \times \mathbb{R}$, on which $\operatorname{arccosh}(\cosh r \cosh b)$ is the hyperbolic hypotenuse law ($\cosh h = \cosh r \cosh b$, legs meeting at a right angle in $\mathbb{H}^2$) while a adds by flat Pythagoras in the $\mathbb{R}$ factor. This exact value is content beyond the projection principle, which yields only the inequality. (ii) On the compactness subtlety: $H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ refers to the intersection with the ambient maximal compact determined by the physical inner product. Abstractly $SO(2,\mathbb{C}) \cong \mathbb{C}^\times$ has maximal compact $U(1)$ after a suitable conjugation; the finiteness is a property of the physical embedding — which is precisely the point: the embedding, not the group, carries the obstruction, and Proposition 2.10 locates the minimal compatible geometry it forces.

2.2 The n-mode statement

The projection principle is dimension-independent, so the variational selection extends verbatim to any number of PT mode pairs.

Theorem 2.14 ($\mathrm{Sp}(2n,\mathbb{C})$ selection principle)**.** *Fix a symplectic splitting of $\mathbb{C}^{2n}$ into n canonical mode planes and let $\mathcal{C}_n \cong \prod_{j=1}^n SL(2,\mathbb{C}) \subset \mathrm{Sp}(2n,\mathbb{C})$ be the mode-preserving subgroup (connected and †-stable, hence compatible), with positive part $\mathbb{P}_{\mathcal{C}_n} \cong (\mathbb{H}^3)^n$. The orthogonal complement of $\mathfrak{c}_{n,\mathfrak{p}}$ in the Hermitian Hamiltonian directions consists exactly of the* inter-mode *couplings (Hermitian Hamiltonian matrices with vanishing mode-diagonal blocks). Consequently, for every inter-mode $\xi = \xi^\dagger$,*

$$\min_{c\in\mathcal{C}_n} \left\| \log\left((e^\xi c)^\dagger e^\xi c\right)\right\|_F = \|2\xi\|_F, \qquad \arg\min = \mathcal{C}_n \cap K, \tag{19}$$

and for any quadratic PT Hamiltonian whose positive normal-form diagonalizer is e^{ξ} with inter-mode ξ, and whose stabilizer is contained in $\mathcal{C}_n$, the positive diagonalizer is the unique minimum-distortion diagonalizer up to matrix-unitary stabilizer elements.

Proof. Write $\xi \in \mathfrak{p}$ in 2×2 blocks $(\xi_{ij})_{i,j=1}^n$ relative to $V = V_1 \oplus \cdots \oplus V_n$. A one-parameter group $e^{t\eta}$ lies in $\mathcal{C}_n$ iff η is block-diagonal, so $\mathfrak{c}_{n,\mathfrak{p}} = \{\xi \in \mathfrak{p} : \xi_{ij} = 0,\ i \neq j\}$ (per-mode Hermitian traceless blocks: Hermiticity and the 2×2 Hamiltonian condition make each ξ_{jj} a traceless Hermitian block). Since $\operatorname{tr}(\xi\eta) = \sum_{ij} \operatorname{tr}(\xi_{ij}\eta_{ji})$ and η is block-diagonal, the pairing sees only $\sum_j \operatorname{tr}(\xi_{jj}\eta_{jj})$, so $\xi \perp \mathfrak{c}_{n,\mathfrak{p}}$ iff every diagonal block ξ_{jj} vanishes: the normal space is exactly the inter-mode couplings, whose paired off-diagonal blocks ξ_{ji} are determined from ξ_{ij} by the Hermitian and symplectic conditions. The dimension identity $n(2n+1) = 3n+2n(n-1)$ is then a check, not a step. Everything else is Theorem 2.9 applied to $\mathcal{C}_n$. □

The theorem ends deliberately at the finite-dimensional Gram-matrix statement. Matrix unitarity is not automatically quantum unitarity: a unitary operator implementing $UVU^{-1} = CV$ in quadrature coordinates ($V = V^\dagger$) must send self-adjoint canonical variables to self-adjoint canonical variables, which requires C to preserve the physical real structure, not merely $C^\dagger C = I$ — a generic element of $\mathrm{Sp}(2n,\mathbb{C}) \cap U(2n)$ has complex coefficients and defines no $*$-automorphism of the real CCR algebra. The Hilbert-space conclusion therefore needs an explicit hypothesis:

Corollary 2.15 (Hilbert-metric constancy under $*$-compatibility)**.** *In the setting of Theorem 2.14, assume in addition that every minimizer $C \in \mathcal{C}_n \cap K$ arising from the physical stabilizer preserves the selected CCR real structure (i.e. is real in quadrature coordinates, hence metaplectically implemented by a genuine unitary). Then all minimizers define the same Hilbert-space metric, by the polar-factor argument of Theorem 2.5. For the PU oscillator this hypothesis holds: the unitary stabilizer elements in the canonical coordinates are the real sign matrices $\{\pm P_1 \pm P_2\} \cong \mathbb{Z}_2^2$ generically, and ordinary real rotations at alignment [1], so the PU metric-constancy conclusion of Theorem 2.5 is unaffected. For a general n-mode system the hypothesis is to be verified per system, alongside the containment of the stabilizer in $\mathcal{C}_n$.*

For the PU oscillator, $n = 2$ and $\xi = \frac{r}{2}B$ is inter-mode; the Bender–Mannheim metric is the case in point. The model-dependent input in any application is thus reduced to a single check: that the positive diagonalizer direction is inter-mode for the chosen splitting, i.e., trace-orthogonal to $\mathfrak{c}_{n,\mathfrak{p}}$ — and by Theorem 2.9(c) this check is not only sufficient but necessary. We emphasize that the canonical-hull computation of Proposition 2.10 is proved here for the two-mode PU stabilizer; for a general n-mode system the containment of the physical stabilizer in $\mathcal{C}_n$, and the structure of its compatible hull across mixed exceptional strata, are hypotheses to be verified per system.

3 The free fourth-order field

3.1 Modewise data and the phase-space Hilbert scale

For $(\Box+m_1^2)(\Box+m_2^2)\phi = 0$, each spatial momentum k carries a PU pair with $\omega_i(k) = \sqrt{k^2+m_i^2}$ and $m_1 > m_2 > 0$. Since $(\omega_1-\omega_2)(\omega_1+\omega_2) = m_1^2 - m_2^2$ identically, the rapidity obeys (3) *exactly*, interpolating between $r(0) = \log\frac{m_1+m_2}{m_1-m_2}$ and $r(k) = 2\log|k| + O(1)$.

Corollary 3.1 (Phase-space Hilbert scale)**.** *In the canonically normalized modewise coordinates, the observable-space metric $M_{\text{obs}}(k) = S_+(k)^\dagger S_+(k)$ has eigenvalues $e^{\pm r(k)}$ with*

$$e^{r(k)} = \frac{\big(\omega_1(k) + \omega_2(k)\big)^2}{m_1^2 - m_2^2} \asymp 1 + k^2, \qquad e^{-r(k)} \asymp (1 + k^2)^{-1}. \tag{20}$$

On the spectral subspaces $\Pi_\pm(k)$ of B, in the canonically normalized mode bundle*, the two spectral directions therefore carry relative weights of orders $+2$ and -2: the induced norm is not uniformly equivalent to the standard one, and no scalar renormalization makes it so. We emphasize that this is a statement relative to the k-dependent normalized trivialization ($d_x(k)d_y(k) = \gamma\omega_1(k)\omega_2(k)$); an invariant Sobolev classification of the* original *field variables would require the pullback $D(k)^\dagger M_{\text{obs}}(k)D(k)$ in a fixed Fourier-space trivialization, since part of the weight may be carried by the normalization itself. We use only the normalized-bundle statement below.*

This is the precise content of the observation that the modewise metric "blows up in the UV". It concerns the classical (phase-space) metric. The quantum state-level comparison is finer, and is computed next.

3.2 The PT ground state, exactly

Fix a mode and work in the normalized coordinates (x, y) with $\omega_1 > \omega_2 > 0$. The positive Hamiltonian $\widetilde{H}'$ has ground state $\phi_0 \propto e^{-\frac{1}{2}(x^2/\omega_2 + \omega_1 y^2)}$. The PT Hamiltonian in these coordinates is the differential operator

$$H'_{\text{PT}} = \tfrac{1}{2}\Big[\tfrac{\omega_1^2+\omega_2^2}{\omega_1\omega_2}\, x^2 + \omega_1\omega_2\, y^2 - \omega_1\omega_2\, \partial_x^2\Big] - x\,\partial_y. \tag{21}$$

Theorem 3.2 (PT ground state)**.** *The unique Gaussian solution of the modewise annihilation equations $\big(\ell_j^\top S_+^{-1} V\big)\psi_0 = 0$ is*

$$\psi_0(x, y) \;\propto\; \exp\Big[-\tfrac{1}{2}\Big(\tfrac{\omega_1+\omega_2}{\omega_1\omega_2}\, x^2 + 2xy + (\omega_1 + \omega_2)\, y^2\Big)\Big], \tag{22}$$

a real *Gaussian with* det(quadratic form) $= \frac{\omega_1^2+\omega_1\omega_2+\omega_2^2}{\omega_1\omega_2} > 0$*: normalizable for all $\omega_1 > \omega_2 > 0$, and $H'_{\text{PT}}\psi_0 = \frac{\omega_1+\omega_2}{2}\,\psi_0$. Moreover ψ_0 is exactly the Dyson pullback of the normal-form vacuum,*

$$\psi_0 = \rho^{-1}\phi_0 \qquad (\textit{up to normalization};\ \rho = e^{-Q/2}), \tag{23}$$

since $\rho^{-1}V\rho = S_+^{-1}V$ transports the annihilation conditions of ϕ_0 into those defining ψ_0 (machine-verified at the Gaussian level). Its identity-embedding Gaussian fidelity *with the two-oscillator vacuum — both wavefunctions normalized in the same auxiliary standard L^2 norm, canonical variables identified by the identity map — and its* auxiliary standard-Fock occupation *relative to it are exactly*

$$f(\omega_1, \omega_2) = \big|\langle\hat{\phi}_0|\hat{\psi}_0\rangle\big|^2 = \frac{4\omega_1\sqrt{\omega_1^2 + \omega_1\omega_2 + \omega_2^2}}{4\omega_1^2 + 3\omega_1\omega_2 + \omega_2^2}, \tag{24}$$

$$\langle N\rangle(\omega_1, \omega_2) = \langle\hat{\psi}_0|(N_1 + N_2)|\hat{\psi}_0\rangle = \frac{\omega_2(\omega_1^2 + \omega_2^2)}{2\omega_1(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)}, \qquad \langle N_1\rangle = \langle N_2\rangle. \tag{25}$$

In particular, along the field dispersion $\omega_i = \omega_i(k)$*,*

$$\lim_{|k|\to\infty} f = \frac{\sqrt{3}}{2} \approx 0.866, \qquad \lim_{|k|\to\infty} \langle N \rangle = \frac{1}{3}, \tag{26}$$

universal constants independent of m_1, m_2, γ*.*

Proof. Solving the two first-order equations for a Gaussian ansatz gives the stated quadratic form (one solution); the eigenvalue equation is a direct substitution. The fidelity is the standard two-Gaussian determinant formula; the occupations follow from covariance algebra, $\langle N_j \rangle = \frac{1}{2} c_j^\top \Sigma c_j$ with Σ the covariance of $|\psi_0|^2$ and c_j the coefficient vectors of $a_j \psi_0$. The limits follow with $\omega_1/\omega_2 \to 1$. (Independent symbolic verification of each step, and a 25-digit quadrature check of the fidelity, are recorded in the repository as verification evidence.) $\square$

Remark 3.3. Two features deserve emphasis. (i) ψ_0 is a smooth, normalizable Gaussian even as $\omega_1 \to \omega_2$: the exceptional-point pathology (divergence of S_+, Jordan structure) lives in the similarity transformation and the excited spectrum, not in the ground state itself. (ii) A naive Bogoliubov estimate would assign the modewise map a squeezing $\sim \sinh^2\big(r(k)/2\big) \sim k^2$ and hence a violently divergent particle content per mode. The exact result (26) is $O(1)$: the diverging rapidity $r(k)$ is carried almost entirely by directions that do not create quanta. The naive estimate fails because $\rho = e^{-Q/2}$ is not $*$-preserving on the original Fock structure — the transformed "annihilators" $\rho a_j \rho^{-1}$ do not satisfy the unitary Bogoliubov normalization ($\alpha\alpha^\dagger - \beta\beta^\dagger = \cosh r \neq 1$) — so Shale–Stinespring-type criteria cannot be applied to ρ directly. Comparing the two *states* as vectors of the same auxiliary L^2 avoids this trap — at the price, made explicit below, that the comparison is then the auxiliary identity-embedding one, not the physical η-completion.

3.3 The physical completion: pointed unitary equivalence

Proposition 3.4 (Pointed unitary equivalence of the physical completions)**.** *Equip the mode space with the physical inner product* $\langle u, v \rangle_{\eta_k} = \langle u, \eta_k v \rangle_{\mathrm{std}}$*. Then* $U_k := \rho_k$ *is unitary from the PT mode space to the standard mode space by construction, and by* (23) *it is* pointed*:*

$$U_k \psi_k = \phi_k, \qquad \langle \phi_k, U_k \psi_k \rangle = 1 \quad \textit{for every mode } k. \tag{27}$$

Consequently the pointed infinite tensor product $U = \bigotimes_k U_k$ *exists with no von Neumann obstruction and maps* $\bigotimes_k \psi_k \mapsto \bigotimes_k \phi_k$*:* with the η-inner product and the Dyson-transported observable algebra $A_{\mathrm{std}} = \rho A_{\mathrm{PT}} \rho^{-1}$, the physical PT completion of the free fourth-order field is globally unitarily equivalent to the positive normal-form Fock completion*,*

$$\omega_{\mathrm{PT}}(A_{\mathrm{PT}}) = \frac{\langle \psi, \eta A_{\mathrm{PT}} \psi \rangle_{\mathrm{std}}}{\langle \psi, \eta \psi \rangle_{\mathrm{std}}} = \langle \phi, A_{\mathrm{std}} \phi \rangle_{\mathrm{std}}. \tag{28}$$

In particular the physical expectation of the transported number observable vanishes: $\langle \rho^{-1} N \rho \rangle_\eta = \langle \phi, N \phi \rangle = 0$*. The unboundedness of* ρ *relative to the standard norm (Corollary 3.1) is real but does not obstruct this equivalence.*

The $\sqrt{3}/2$ and $1/3$ of Theorem 3.2 are therefore *not* the physical overlap and particle content of the PT vacuum; they quantify the mismatch of two auxiliary Gaussian complex structures when the canonical variables are identified by the identity map. That mismatch is itself an exact and structured obstruction:

3.4 The auxiliary identity-embedding obstruction

Theorem 3.5 (Auxiliary identity-embedding obstruction)**.** *Let $m_1 > m_2 > 0$, $d \geq 1$, and consider the free fourth-order field in a box of volume V with momentum cutoff Λ. Regard the normalized wavefunctions $\hat{\psi}_0(k)$ and $\hat{\phi}_0(k)$ as vectors of one auxiliary Schrödinger representation — same standard L^2 inner product, same standard CCR involution, canonical variables identified by the identity map — and let $\Psi_{\mathrm{aux}} = \bigotimes_k \hat{\psi}_0(k)$, $\Omega_{\mathrm{aux}} = \bigotimes_k \hat{\phi}_0(k)$ be the two auxiliary product states. Then (independent modes labelled by the $(k, -k)$ pairs of the real field; a different mode convention changes the prefactor below by a fixed factor):*

1. *the auxiliary overlap has the exact leading asymptotics*

$$-\log \left|\langle \Omega_{\mathrm{aux}} | \Psi_{\mathrm{aux}} \rangle\right|^2 = \left[\log \tfrac{2}{\sqrt{3}}\right] \frac{\mathrm{vol}(B_d)}{(2\pi)^d} \, V\Lambda^d + o\big(V\Lambda^d\big), \tag{29}$$

   with universal leading coefficient (since $f(k) \to \sqrt{3}/2$ and $-\frac{1}{V} \log |\langle \Omega, \Psi \rangle|^2 = -\int_{|k|\leq\Lambda} \frac{d^dk}{(2\pi)^d} \log f(k)$);

2. *the auxiliary standard-Fock occupation density diverges in the UV: $\frac{1}{V}\langle N_{\mathrm{tot}}\rangle = \int^{\Lambda} \frac{d^dk}{(2\pi)^d} \langle N\rangle(k) \sim \frac{1}{3}\int^{\Lambda} \frac{d^dk}{(2\pi)^d} \to \infty$ as $\Lambda \to \infty$, in* every *dimension $d \geq 1$;*

3. *in the thermodynamic/continuum limit the incomplete-tensor-product classes of Ψ_{aux} and Ω_{aux} differ ($\sum_k (1 - |\langle \hat{\psi}_0 | \hat{\phi}_0 \rangle|) = \infty$), so the standard-$L^2$ Gaussian state defined by the PT eigenfunction is disjoint from the standard two-field vacuum state* under the identity embedding of the auxiliary canonical variables*: no density matrix in the auxiliary Fock representation reproduces it, and vice versa.*

This theorem compares two vector states on one fixed auxiliary standard CCR algebra. It does not *compare the physical η-completion with the transported normal-form completion — those are equivalent by Proposition 3.4 — and its conclusion may not be called disjointness of the physical PT representation.*

Proof. (1) and (2) are immediate from Theorem 3.2: $f(k)$ and $\langle N\rangle(k)$ are continuous in k with the universal limits (26); the Riemann-sum limit of $-\log f$ gives the stated coefficient. (3) is von Neumann's equivalence criterion for incomplete tensor products [9] applied to the two product vectors, combined with purity: inequivalent pure product states generate disjoint representations. □

Remark 3.6 (What this does and does not say). The two results together resolve what would otherwise be a contradiction. The physical Hilbert space, defined as the completion of the modewise-unitarized theory, is naturally identified with the Fock space of the free pair (m_1, m_2), globally and unitarily (Proposition 3.4); on it the dynamics is unitary, exactly as argued by Bender and Mannheim modewise — and this equivalence is *pointed*, with per-mode overlap exactly one. What fails is something different: the identity embedding of the auxiliary canonical variables does not extend to a normal equivalence of the two auxiliary quasifree states (Theorem 3.5), with an $O(1)$ discrepancy per UV mode. The obstruction is the incompatibility between the *identity embedding of canonical variables* and the *Dyson-transported physical involution* — an involution-level mismatch, not a Hilbert-space inequivalence, in line with the three-level distinction (complex canonical dynamics $\neq$ real form/involution $\neq$ Hilbert completion) developed for the field theory in the companion papers. Practical consequences survive in sharpened form: any interacting construction must declare which embedding of the canonical variables it uses; formal manipulations that insert $\eta = e^{-Q}$ as if it were a bounded operator on the covariant Fock space are unsupported (by Corollary 3.1 the metric is unbounded with unbounded inverse); and PT-based unitarity claims for quadratic gravity [10] require an explicit statement of involution and completion, not only of the modewise similarity.

4 Discussion

The variational principle. Theorem 2.5 upgrades the uniqueness theorem of [1]: the Bender–Mannheim metric is not merely the unique positive point of the diagonalizer coset but its unique distortion minimizer, with the exact excess $4[a^2 + \operatorname{arccosh}^2(\cosh r \cosh b)] - 4r^2$ in the Gram rapidities of the stabilizer. Section 2.1 shows this is an instance of an orthogonal Cartan-norm projection principle along †-closed subgroups — classical in spirit [6, 7], here proved in if-and-only-if form with the minimal compatible hull identified canonically (Proposition 2.10) — while the elementary proof supplies the minimizing value in closed form, which the abstract theory does not. The n-mode version (Theorem 2.14) makes the portability to general quadratic PT systems a theorem rather than a claim.

A scope distinction now matters: minimum distortion is *free geometric optimality*, not *interacting dynamical stability*. Among admissible free positive diagonalizers the Bender–Mannheim metric is the geometrically least distorting; the theorem does not make it the most stable under interaction, uniquely physical, or guaranteed to survive scattering. Indeed the companion interaction paper [2] proves that the order-by-order deformation of this metric is obstructed at rational resonances and on open scattering shells. That result sharpens rather than weakens the present one: the obstruction strikes the *optimal* free positive metric, so the interacting failure cannot be dismissed as the consequence of an arbitrary or badly chosen metric.

The three levels, again. The field-theoretic analysis separates three statements that are easily conflated. The *phase-space* metric degenerates in the UV exactly as the rapidity formula (3) dictates ($e^{\pm r(k)} \asymp k^{\pm 2}$ in the normalized mode bundle); this is a statement about observables. The *physical completion* is globally unitarily equivalent to the normal-form Fock completion, by a pointed unitary with per-mode overlap one (Proposition 3.4); this is a statement about Hilbert completions. The *auxiliary identity-embedding* mismatch saturates at universal constants ($\sqrt{3}/2$, $1/3$) and its product states are disjoint (Theorem 3.5); this is a statement about the incompatibility of the identity embedding with the Dyson-transported involution. The revised central conclusion is therefore:

> *The Dyson map gives an exact pointed unitary equivalence between the physical PT completion and the positive normal-form Fock completion. However, if the two Gaussian vacua are compared through the identity embedding of the auxiliary canonical variables and the standard involution, their standard-CCR product states are disjoint, with universal ultraviolet fidelity and occupation defects.*

This aligns with the three-level structure — complex canonical dynamics $\neq$ real form/involution $\neq$ Hilbert completion — used in the companion papers. All quantities are exact consequences of the same modewise data. The constants are universal because the UV limit $\omega_1/\omega_2 \to 1$

erases all mass and coupling dependence — remarkably, the same limit in which the finite-dimensional theory degenerates (the exceptional point) is the limit in which the ground-state data become featureless.

Outlook. Three natural continuations: (i) sharpen Theorem 2.14 into a classification: for which quadratic PT Hamiltonians is the positive diagonalizer direction inter-mode for some canonical splitting (by Theorem 2.9(c) this is necessary and sufficient), and what replaces the selection principle when it fails; (ii) determine whether the auxiliary identity-embedding obstruction survives interactions in low dimensions, where the free-theory defect is only the universal 1/3 per mode (a wave-function renormalization of the modewise map, $\rho_k \to \rho_k Z_k$ with Z_k unitary, cannot change class (3), but a k-dependent redefinition of the mode split might); and (iii) make contact with the Krein-space formulation of indefinite-metric QFT, where the covariant representation lives, to phrase the representation choice intrinsically. We leave these to future work.

References

[1] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper and verification repository, 2026.

[2] *Interaction obstructions, resonant PT breaking, and doubled Jordan symmetry in fourth-order theories*, companion paper (Paper V), 2026.

[3] C. M. Bender and P. D. Mannheim, Phys. Rev. Lett. **100** (2008) 110402.

[4] C. M. Bender and P. D. Mannheim, Phys. Rev. D **78** (2008) 025022.

[5] R. Bhatia, *Positive Definite Matrices*, Princeton University Press, 2007.

[6] G. D. Mostow, *Self-adjoint groups*, Ann. of Math. **62** (1955) 44.

[7] G. Kempf and L. Ness, *The length of vectors in representation spaces*, in: Algebraic Geometry, Lecture Notes in Math. **732**, Springer, 1979, 233.

[8] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Springer, 1999.

[9] J. von Neumann, *On infinite direct products*, Compos. Math. **6** (1939) 1.

[10] J. Kubo and J. Kuntz, *Unitarity through PT symmetry in quantum quadratic gravity*, arXiv:2410.08278.